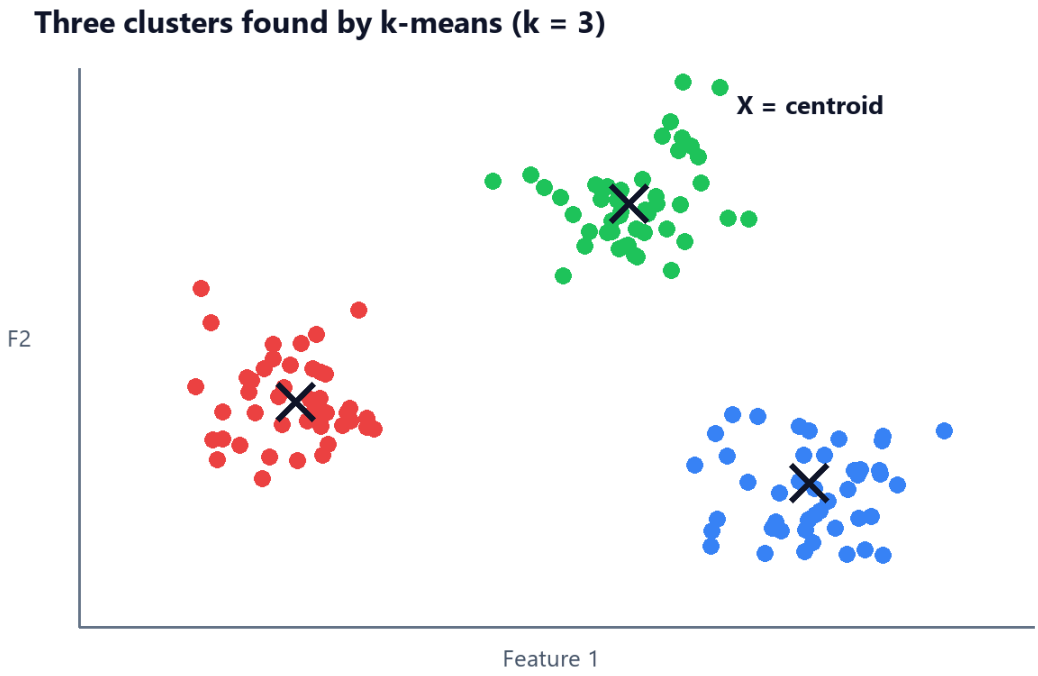
k-means: Lecture Notes

These notes accompany Lecture 4 (Introduction to Clustering).

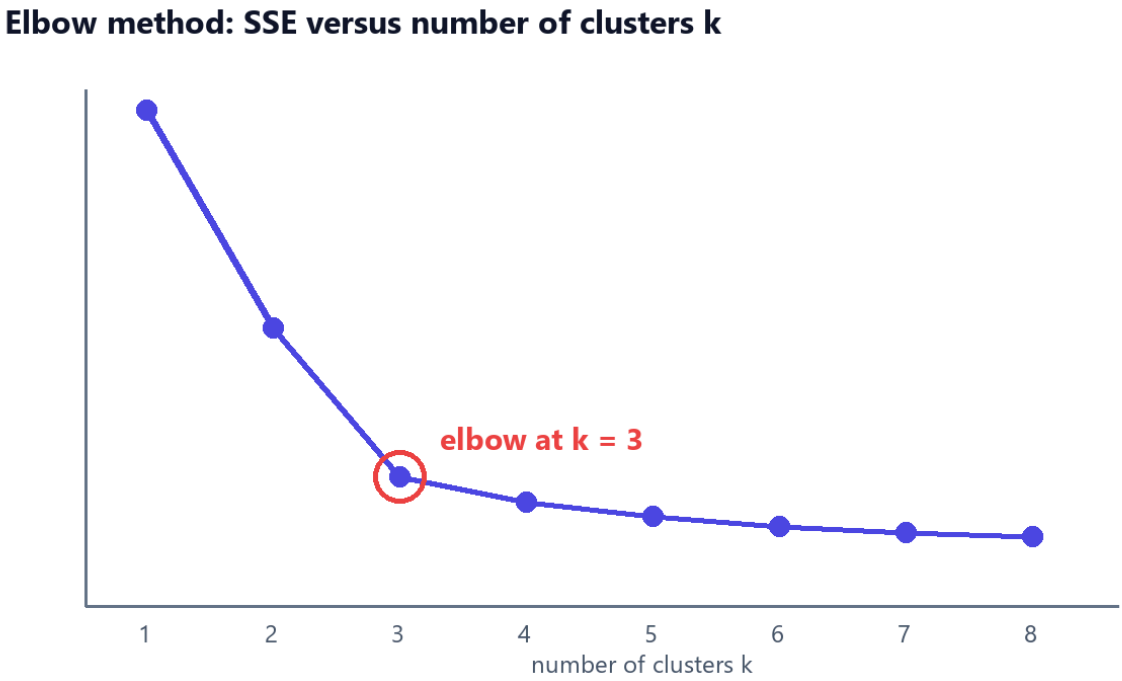
k-means is a partitioning method. Given a number of clusters K chosen in advance, it searches for the partition that minimizes the total within-cluster sum of squared errors (SSE): the sum, over all clusters, of the squared Euclidean distances between each observation and the centroid of its cluster.

Algorithm: (1) choose K initial centroids; (2) assign each observation to the nearest centroid; (3) recompute each centroid as the mean of its assigned observations; (4) repeat steps 2 and 3 until no observation changes cluster. Each iteration can only decrease the SSE, so the algorithm always converges, but it may converge to a local minimum.

Worked example: with observations 2, 4, 10, 12, 20 and 30 and initial centroids 4 and 12, the first assignment is {2, 4} and {10, 12, 20, 30}. The new centroids are 3 and 18. After re-assignment the clusters are {2, 4, 10} and {12, 20, 30} with centroids 5.33 and 20.67, and the algorithm stops after one more iteration because the assignment no longer changes.



Choosing the number of clusters

The elbow method runs k-means for K = 1, 2, 3, ... and plots the SSE for each K. SSE always decreases as K grows, so the best K is not the minimum SSE. Instead we look for the 'elbow': the point after which adding another cluster gives only a small decrease in SSE. In the figure below the elbow is at K = 3.

Practical advice: standardize variables before clustering, run k-means several times with different initial centroids (or use k-means++ initialization) and keep the solution with the lowest SSE, and remember that k-means is sensitive to outliers because centroids are means.